

Module 1: Morphology & Language Models

Linguistic Hierarchy, Tokenization (BPE), N-grams, Perplexity & Kneser–Ney Smoothing

Module I: Introduction to NLP & Preprocessing — Complete 12-Topic Tracker

Topic 1: Introduction to NLP (Scientific vs. Engineering Goals & Ambiguity)	Topic 2: Real-World NLP Applications (Search, Healthcare, Finance, Chatbots)
Topic 3: Morphology (Morphemes, Free/Bound, Inflectional vs. Derivational)	Topic 4: Syntax (Grammatical Phrase Structures & Parse Trees)
Topic 5: Semantics (Literal Meaning & Semantic Role Labeling)	Topic 6: Pragmatics (Contextual Meaning: S-S-P Paradigm)
Topic 7: Tokenization (Word, Sentence, Subword BPE & Character)	Topic 8: Lemmatization (Base Lemma vs. Porter Stemming)
Topic 9: Stop-Word Removal (Tradeoffs & Sentiment Preservations)	Topic 10: Normalization (Case Folding, Unicode & Contractions) [UPLOADED PYQ]
Topic 11: Regular Expressions (Pattern Matching & Information Extraction)	Topic 12: Language Models Overview (Sequence Probability Distributions)

Topic 1 & 2: Introduction to NLP & The Fundamental Ambiguity Challenge [UPLOADED PYQ]

Natural Language Processing (NLP) sits at the intersection of Artificial Intelligence and Computational Linguistics, pursuing two fundamental objectives:

Scientific Goal: Formulating computational models of human cognitive language understanding and production.

Engineering Goal: Developing robust, scalable software systems capable of processing human text and speech for practical real-world utility.

[UPLOADED PYQ — CS633] Why is Ambiguity the Core Challenge of NLP?

Natural language is inherently ambiguous across all linguistic tiers:

1. **Lexical Ambiguity:** A single word possesses multiple distinct meanings (e.g., "bank" → financial institution vs. river slope).
2. **Syntactic (Structural) Ambiguity:** A sentence can generate multiple valid parse trees (e.g., "I saw the man with the telescope" → who holds the telescope?).
3. **Semantic Ambiguity:** A sentence has multiple literal interpretations (e.g., "The chicken is ready to eat" → is the chicken hungry or cooked?).
4. **Pragmatic Ambiguity:** Meaning depends strictly on conversational context and real-world intent (e.g., "Can you pass the salt?" → an action request, not an inquiry about physical ability).

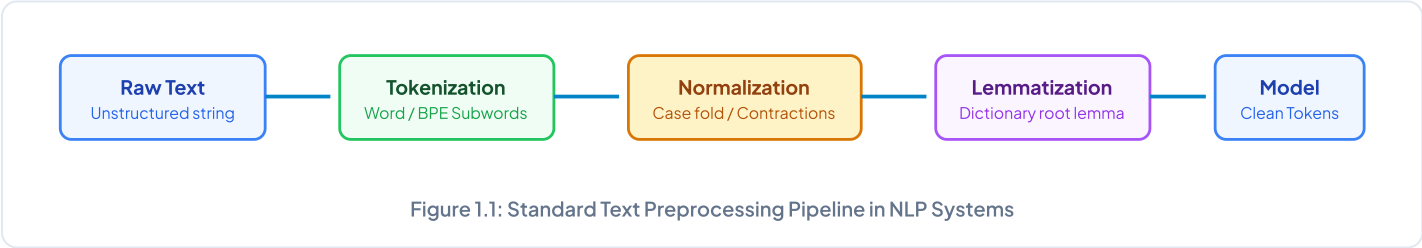
Topic 3 – 6: The 4 Core Levels of Linguistic Analysis

Linguistic Level	Core Scope & Analysis Mechanism	Representative Example
1. Morphology	Study of the internal structure of words and their minimal meaning-bearing units (Morphemes). • <i>Free</i>: Can stand alone ("happy"). • <i>Bound</i>: Prefixes/Suffixes ("un-", "-ness"). • <i>Inflectional</i>: Changes grammar form ("walk" → "walked"). • <i>Derivational</i>: Creates new words/parts of speech ("happy" → "unhappiness").	`unhappiness` = `un-` (prefix) + `happy` (root) + `-ness` (suffix).
2. Syntax	Formal grammatical rules governing how words combine into phrases and sentences (Context-Free Grammars).	$S \rightarrow NP VP$ (e.g., "The student [NP] solved the problem [VP]").
3. Semantics	The study of literal, context-independent linguistic meaning and predicate-argument relationships (Semantic Roles: Agent, Action, Patient).	"The dog chased the cat" → Agent = dog, Action = chase, Patient = cat.
4. Pragmatics	Meaning in real-world communicative context, discourse intent, and speaker implicature.	"It is freezing in here!" ⇒ Pragmatic command to close the window.

 **Memory Hook: S-S-P Paradigm**

Syntax (Structure) → Semantics (Sense / Meaning) → Pragmatics (Situation / Context)

Topic 7 – 10: Text Preprocessing & Normalization Pipeline [UPLOADED PYQ]



Feature	Stemming (e.g., Porter Stemmer)	Lemmatization (WordNet Lemmatizer) [UPLOADED PYQ]
Core Technique	Crude heuristic rule-based chopping of prefixes and suffixes.	Full morphological vocabulary and dictionary lookup analysis.
Output Validity	Often produces non-words (e.g., "running" → "run", "studies" → "studi").	Guaranteed valid dictionary base form (e.g., "better" → "good", "mice" → "mouse").
Context Awareness	Zero Part-of-Speech context awareness.	Uses POS tags to resolve ambiguity ("meeting" as Verb → "meet", as Noun → "meeting").

Topic 11 & 12: Regular Expressions & Language Models Overview

Language Model Formal Definition: A **Language Model (LM)** computes the joint probability distribution over any arbitrary sequence of n words $W = \langle w_1, w_2, \dots, w_n \rangle$:

$$P(w_1, w_2, \dots, w_n) = \prod_{i=1}^n P(w_i \mid w_1, w_2, \dots, w_{i-1})$$

M1 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ — CS633] Explain the 5 sequential phases of NLP with an end-to-end example. (10 Marks)

Given sentence: "The intelligent student solved the complex problem."

1. **Morphological Analysis:** Breaks tokens into morphemes ("solve" + "-ed", "complex").
2. **Syntactic Analysis (Parsing):** Builds hierarchical phrase parse tree ($S \rightarrow NP VP$).
3. **Semantic Analysis:** Assigns literal meaning (*Agent* = student, *Action* = solve, *Object* = problem).
4. **Discourse Integration:** Links pronouns and entities across surrounding sentences.
5. **Pragmatic Analysis:** Interprets the speaker's true intent within conversational context.